

SIH26143

Satellite Oil-Spill Detection

Deep Technical Research Report

Sentinel-1 SAR • Semantic Segmentation • Environmental Context • AIS Correlation

Purpose

A technically deep but hackathon-oriented research report for the satellite/oil-spill module of Smart India Hackathon 2026 Problem Statement SIH26143. The goal is to move from physical understanding and verified public data to a working segmentation baseline within roughly one day, while identifying the research-grade extensions that can make the prototype more credible.

Decision	Recommendation
Primary sensor	Sentinel-1 C-band SAR, especially IW GRD products
Primary dataset	Deep-SAR Oil Spill (SOS) - Sentinel-1 subset
Baseline	U-Net with BCE + Dice loss
Stretch model	MobileNetV2-DeepLabV3+
Critical context	Wind speed/direction, acquisition geometry, sea state
Main failure mode	Look-alikes / dark non-oil signatures
SIH differentiator	Spill geometry + environmental context + AIS trajectory attribution

Research status: Sources and dataset claims were checked against current public pages where available. Dataset sizes, model results, and access details are reported as the cited sources describe them; they should not be interpreted as a guarantee of field performance.

Executive Summary

SIH26143 asks for an automated pipeline that detects and characterises marine oil spills from satellite imagery, combines oceanographic and meteorological information to reason about slick movement/origin, and correlates the inferred origin window with historical AIS traffic to help identify the responsible vessel. A current public description of the problem explicitly frames the challenge around detection, characterisation, environmental modelling, and AIS attribution. [1]

The key remote-sensing fact is simple: SAR does not see petroleum directly. It sees changes in the ocean surface roughness. Wind-generated capillary/gravity-capillary waves create radar backscatter. A surface oil film damps those short waves, reducing backscatter, so a slick often appears as a dark region. EMSA stresses that the same mechanism is produced by natural films, current patterns, low-wind zones, rain and other phenomena; therefore SAR detections are properly treated as possible oil spills, not chemically definitive identifications. [5,6]

For the one-day prototype, the best engineering decision is to avoid downloading and preprocessing a massive archive of raw Sentinel-1 scenes. The Deep-SAR Oil Spill (SOS) dataset is a manageable benchmark containing 8,070 labelled 256×256 samples derived from 21 original SAR scenes, including a Sentinel-1A VV subset from the Persian Gulf. Multiple published studies use it for segmentation experiments.

[7,8]

A plain U-Net is the correct baseline: easy to train, easy to visualise, easy to explain, and directly suited to pixel-wise segmentation. A stronger stretch model is MobileNetV2-DeepLabV3+, which has already been evaluated on SOS and offers a useful accuracy/complexity trade-off. [8]

The hardest technical issue is not obtaining a high validation IoU on a benchmark. It is generalisation to real scenes, especially dark look-alikes and environmental conditions outside the training distribution. Recent datasets and studies increasingly separate oil slicks from look-alikes and explicitly model contextual information. [12,19]

Recommended prototype: SOS Sentinel-1 subset → normalisation → U-Net → BCE+Dice → binary mask → connected components / polygon → area & centroid → wind/context check → AIS search window.

Contents

#	Section
1	Problem definition and SIH system scope
2	Remote-sensing physics of oil-spill detection
3	Why SAR and why Sentinel-1
4	Sentinel-1 data products and acquisition geometry
5	Environmental conditions and wind-speed physics
6	What an oil slick looks like in SAR
7	False positives and look-alikes
8	Datasets: verified comparison
9	SOS dataset deep dive
10	Alternative Sentinel-1 datasets
11	Existing detection and segmentation methods
12	Model architecture analysis
13	Recommended baseline
14	Preprocessing and training pipeline
15	Evaluation metrics and validation design
16	From mask to spill characterisation
17	Environmental backtracking and drift
18	AIS correlation and vessel attribution
19	End-to-end SIH architecture
20	Prototype implementation plan
21	Technical risks and mitigation
22	Research-grade extensions
23	Final decisions for the hackathon
24	References and verified links

1. Problem Definition and SIH System Scope

The public SIH26143 description frames the challenge as using satellite imagery, including SAR and EO imagery, together with AIS data to detect oil spills and identify the vessel responsible. It also envisages calculating spill geometry/age where feasible, using oceanographic and meteorological data to trace the slick toward an origin, predicting future movement, and reconstructing vessel traffic around the inferred origin window. [1]

This means the satellite module should not be designed as an isolated classifier. It should produce structured outputs that downstream modules can use.

What your module should expose
Satellite module output

1. Spill probability mask
2. Spill polygon(s)
3. Area / perimeter / centroid
4. Confidence score
5. Acquisition timestamp
6. Image geometry / orbit metadata
7. Environmental-quality flags
8. Candidate origin region / uncertainty

The cleanest division of responsibilities is: your module detects and characterises the slick; the environmental module reasons about transport/drift; the AIS module searches vessel trajectories in a time-space window around the inferred source. Keeping these interfaces explicit prevents the prototype from becoming one giant model.

2. Remote-Sensing Physics of Oil-Spill Detection

2.1 The radar backscatter mechanism

A SAR sensor transmits microwave energy and records the returned signal. Over the ocean, the returned signal is strongly influenced by small-scale surface roughness. Wind creates short capillary and gravity-capillary waves. At the appropriate geometry, these waves produce strong Bragg scattering. EMSA describes the operational principle directly: radar detects phenomena that suppress capillary waves; oil reduces the returned signal and therefore appears dark relative to surrounding sea clutter.

[5]

Wind → short surface waves → stronger radar backscatter → brighter ocean
Oil film → damping of short waves → weaker backscatter → darker slick

Important: radar is measuring a surface-scattering response, not oil chemistry.

2.2 Why a thin film can matter

Oil can form a relatively thin surface film yet still modify the small-scale wave spectrum that contributes strongly to microwave scattering. The practical consequence is that a slick can become visible even when its physical thickness is far below the spatial resolution of a pixel. The image therefore represents an integrated electromagnetic response of the surface, not a direct measurement of oil thickness.

2.3 Why the signature is ambiguous

The same electromagnetic mechanism can occur without petroleum. A calm patch, natural surfactant/biogenic film, rain-affected area, internal-wave structure, current boundary, ice-related phenomenon, or other surface condition may also damp short waves. Consequently, the segmentation problem is fundamentally a look-alike discrimination problem as well as an oil segmentation problem.

[5,6]

2.4 A useful conceptual signal model

Observed SAR intensity $\approx$ function(surface roughness, wind, incidence angle, polarization, wavelength, oil properties, geometry, processing noise)

Oil detection = identify coherent spatial regions whose backscatter reduction is more consistent with an oil slick than with competing explanations.

This explains why adding context is scientifically justified rather than merely a machine-learning trick.

3. Why SAR, and Why Sentinel-1?

3.1 SAR versus optical imagery

Property	Optical	SAR
Illumination	Needs reflected sunlight	Active sensor; day and night
Clouds	Can block or contaminate observations	Microwave radar largely penetrates cloud cover
Oil signature	Spectral/colour/texture effects can help	Surface roughness/backscatter suppression
Night operations	Not suitable for ordinary passive imaging	Suitable
Primary weakness	Clouds, haze, illumination, sea-state dependence	Look-alikes, wind dependence, speckle, geometry
SIH role	Useful secondary evidence	Primary detection modality

Copernicus describes Sentinel-1 SAR as capable of day/night and all-weather monitoring and explicitly lists oil-spill monitoring among mission applications. [3,2]

3.2 Why Sentinel-1 is especially suitable for a student prototype

- Open and free access through the Copernicus Data Space Ecosystem. [4,3]
- Long historical archive beginning in 2014, with global GRD coverage. [4,3]
- C-band SAR at 5.405 GHz, directly relevant to ocean-surface roughness monitoring. [2]
- IW mode provides a 250 km swath and approximately 5×20 m spatial resolution, making it practical for wide-area maritime surveillance. [2]
- Public oil-spill datasets and published deep-learning studies already exist, lowering the barrier to a hackathon prototype.

3.3 Current constellation context

Sentinel-1A ended operations in June 2026 according to ESA's current mission facts page; Sentinel-1C and Sentinel-1D are now the operational pair, restoring a six-day equatorial repeat cycle for the two-satellite constellation. The historical datasets used for your prototype can still contain Sentinel-1A imagery, while new acquisitions should be selected using the current constellation. [2]

4. Sentinel-1 Data Products and Acquisition Geometry

4.1 GRD versus SLC

Product	What it contains	Use for SIH
GRD	Detected amplitude/backscatter imagery, ground-range processed	Recommended starting point
SLC	Complex data with amplitude and phase	Research-grade; unnecessary for baseline
RAW/L0	Unprocessed radar measurements	Not appropriate for one-day prototype

For oil-spill semantic segmentation, GRD is the sensible starting point because the model normally consumes backscatter/intensity rather than phase. Copernicus provides Sentinel-1 GRD worldwide from October 2014 onward and makes Sentinel products freely available. ^[4,3]

4.2 Polarisation

Sentinel-1 supports VV, VH and other polarisation configurations depending on acquisition mode. The SOS Sentinel-1 subset used in published work is VV-only. Current Sentinel-1 IW GRD data can commonly provide VV and VH, and a future model can exploit both channels if the training data support it.

^[4,7]

VV is attractive for the baseline because it keeps the input one-channel and aligns directly with the SOS Sentinel subset. A two-channel VV+VH model is a sensible extension, but it creates a domain mismatch if the training benchmark contains only VV.

4.3 Incidence angle and viewing geometry

Backscatter depends on incidence angle. The same sea state can produce different intensities at different viewing geometries. This matters because a model trained on narrow geographic/acquisition distributions may accidentally learn scene-specific intensity patterns. A robust system should therefore retain acquisition metadata and, for larger-scale deployment, consider angle normalisation or geometry-aware training.

4.4 Speckle

SAR is coherent imaging, so speckle is inherent. Speckle creates granular intensity variation that can fragment boundaries and confuse small targets. For a pretrained benchmark, avoid aggressive filtering unless the benchmark documentation expects it. For raw GRD workflows, speckle handling can be introduced carefully; excessive smoothing can erase small slick boundaries.

5. Environmental Conditions and Wind-Speed Physics

Wind is arguably the most important environmental variable for interpreting a dark SAR signature over the sea. EMSA reports that wind speeds below roughly 2-3 m/s can mask the oil-film damping effect because the sea is already too smooth, while speeds above about 15 m/s reduce detection capability.

[5]

Approx. wind regime	Physical situation	Detection implication
< 2-3 m/s	Very smooth sea surface	High false-positive risk; oil contrast may disappear
~3-10 m/s	Useful roughness contrast	Generally favourable operating regime
~10-15 m/s	Increasingly rough / energetic sea	Contrast becomes less reliable
> ~15 m/s	Very rough sea	Detection capability reduced; slick may be difficult to delineate

These ranges are not hard physical thresholds. They are operational guidance. Wind direction, wave age, fetch, current, oil type, slick age, incidence angle and sensor characteristics all modify the observed contrast. ESA training material similarly notes that the contrast depends on wave height, wind speed, oil type, wavelength, polarisation and incidence angle. [6]

5.1 Why wind should become a model feature

A pure image model has to infer environmental context indirectly from texture. If an external wind estimate is available, the system can explicitly flag scenes where the SAR interpretation is unreliable. For a hackathon, even a simple rule-based quality flag can add substantial scientific credibility.

SAR image → U-Net → candidate slick
Wind speed/direction → quality assessment
If very low or very high wind → reduce confidence / flag for review
Else → proceed to geometric and AIS correlation

5.2 Where to get wind

For a research extension, reanalysis products such as ERA5 provide hourly wind variables on a regular grid. The Copernicus Climate Change Service documentation exposes 100 m wind components and a 0.25° grid in its ERA5 application documentation. [22]

API-derived estimate is sufficient; do not build a full ocean model before the baseline works.

6. What an Oil Slick Looks Like in SAR

Typical appearance is a dark, often elongated, irregular or filamentary region. Geometry varies with spill age, wind/current forcing, source type and transport. A recent release of the DARTIS dataset is specifically designed around oil slicks, look-alikes and other remarkable SAR signatures in the Eastern Mediterranean, reinforcing that shape alone is not enough. ^[12]

Useful visual cues

- Darkness relative to surrounding sea rather than absolute brightness alone.
- Spatial continuity: a slick often forms a coherent region rather than isolated dark pixels.
- Elongation and filamentary structure caused by wind/current transport.
- Edges that may be soft, irregular, or broken by changing sea state.
- Location in open water and plausible relationship to maritime traffic.
- Temporal consistency when multiple acquisitions exist.

What the model should not learn

- Absolute grayscale value = oil.
- One particular coastline/background texture = oil.
- One geographic region = oil.
- One acquisition date or orbit = oil.
- Any elongated dark line = oil, because wakes and oceanic features can share that geometry.

A strong prototype therefore needs a confidence layer. The segmentation mask should be treated as a candidate detection, then checked against environmental and maritime evidence.

7. False Positives and Look-Alikes

7.1 Main classes

False positive	SAR mechanism	Why dangerous	Mitigation
Low-wind area	Naturally smooth surface	Can look almost identical to an oil slick	Wind threshold / contextual feature
Natural biogenic film	Surfactant damping	Same physical darkening mechanism	Look-alike training / context
Ship wake	Wake-induced roughness pattern	Linear structures can resemble slick boundaries	AIS + geometry + object context
Rain cell	Changes sea-surface scattering	Can create dark/bright patches	Weather context
Internal waves	Surface roughness modulation	Large coherent dark structures	Context + morphology
Current fronts	Surface convergence/divergence	Filamentary structures	Oceanographic context
Ice / grease ice	Surface scattering changes	Dark signatures in cold regions	Geographic/temperature context
Sensor artifact	Processing / acquisition effects	Can produce anomalous intensity	Calibration / quality flags
Land/coast	Different scattering regime	Strong boundaries confuse segmentation	Land-sea mask

EMSA explicitly identifies natural films, current patterns, ice and biologically associated surface slicks as dark SAR patterns and emphasises the dependence on wind. [5]

7.2 The strongest mitigation: train against look-alikes

A binary dataset containing only oil and generic sea can produce a model that learns the shortcut 'dark region = oil'. A more realistic benchmark separates oil from look-alikes. The M4D line of work, for example, defined foreground classes for oil and look-alikes with background pixels, while the DARTIS dataset explicitly contains oil slicks, look-alikes and other remarkable SAR signatures. [14,12]

For SIH, this suggests a staged approach: train the first U-Net on SOS to get the pipeline working, then add a small look-alike rejection/classification layer if time allows.

8. Verified Dataset Comparison

The table below separates datasets that are useful for segmentation from datasets that are mainly useful for classification or future robustness work.

Dataset	Sensor	Size / samples	Labels	Format	Practicality
Deep-SAR SOS	Sentinel-1A VV + ALOS PALSAR HH	8,070 × 256 ² labelled samples	Pixel masks	Image + mask folders	Best for tomorrow
Trujillo Part I	Sentinel-1 VV+VH	1,200 oil images; 2048 ² ×2	2048 ² masks	TIFF + masks	Excellent but 40.7 GB
Trujillo Part II/III	Sentinel-1	Look-alikes/no-oil + 450 test images	Pixel masks	TIFF	Excellent future robustness
CSIRO Oil Spill	Sentinel-1	5,630 images	Binary image labels	400×400 grayscale	Good classification fallback
DARTIS 2019	Sentinel-1	3,655 patches	Oil + no-oil/look-alike annotations	JPG + Pascal VOC XML	Strong look-alike research
M4D Oil Spill Detection	Sentinel-1	1,110 images	5 classes incl. oil/look-alike/ship/land/sea	Image + masks	Strong multi-class direction

8.1 SOS dataset - detailed verification

The SOS dataset is documented in peer-reviewed work as consisting of 14 PALSAR images from the Gulf of Mexico and 7 Sentinel-1A images from the Persian Gulf. The original scenes were cropped/augmented, manually labelled and converted into 8,070 256×256 labelled samples. The Sentinel-1 training portion contains 3,354 samples, with 839 test samples in the commonly reported split.

[7,8]

A particularly important new finding for this project is that a refined SOS dataset was released on Zenodo in 2025. Its creators manually corrected substantial portions of the training and validation masks, reporting revisions to about 38% of training masks and 50% of validation masks. The release contains approximately 1.1 GB of images plus 28.3 MB of masks. ^[11]

This refined version is worth considering if the download is practical: it addresses label noise in the original benchmark, which can otherwise cap model performance or teach incorrect boundaries.

9. SOS Dataset Deep Dive

Source structure and provenance

- 14 ALOS PALSAR scenes from the Gulf of Mexico / Deepwater Horizon region.
- 7 Sentinel-1A scenes from the Persian Gulf.
- Original scenes were cropped and augmented by rotation/noise/cropping.
- Final benchmark: 8,070 256×256 labelled samples.
- Published work reports an 80:20 train/test split with separate counts for the PALSAR and Sentinel subsets. [7,8]

The dataset is useful because the engineering unit is already a small tensor-like image rather than a multi-gigabyte raw satellite product. That makes it suitable for a rapid U-Net proof of concept.

But there is a major methodological warning: many 256×256 samples originate from a small number of original scenes. Randomly splitting patches can therefore create spatial/scene leakage, where training and validation patches share the same underlying scene characteristics. This can inflate apparent generalisation.

For your internal hackathon this is acceptable as a prototype benchmark, but the report/demo should say: benchmark performance is not equivalent to global operational performance.

10. Alternative Dataset Deep Dive

10.1 Trujillo et al. Sentinel-1 dataset

Zenodo Part I contains 1,200 Sentinel-1 Sigma0 images in dB, each 2048×2048×2 using VV and VH, plus 1,200 corresponding masks. The files are TIFF. The compressed image archive is 40.7 GB, making it a poor choice for a one-night build but an excellent future dataset because it uses larger, georeferenced Sentinel-1 inputs. [10]

Part II adds 685 no-oil and 685 look-alike images with ground truth, while Part III provides a structured test set containing oil, no-oil and look-alike examples. This is much closer to the real discrimination problem than an oil-only benchmark. [10]

10.2 DARTIS 2019

The DARTIS dataset covers the Eastern Mediterranean Sea in 2019 and contains oil slicks, look-alikes and other remarkable SAR signatures. Its image patches are normalised to 0-255 and stored as 8-bit JPG, with oil annotations in Pascal VOC XML. A data table links patches to Sentinel-1 product IDs, allowing the original satellite products to be retrieved from Copernicus. Public code is also associated with the dataset.

[12]

10.3 M4D

M4D research uses a Sentinel-1 dataset with five classes: oil spill, look-alike, ship, land and sea. A 2026 paper describes a 1,110-image benchmark and reports multi-class segmentation results using a boundary-aware dual-attention U-Net. This is highly relevant to a future model because it directly attacks the look-alike problem. [15]

11. Existing Detection and Segmentation Methods

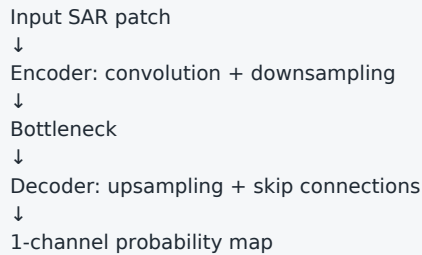
Method / work	Dataset	Core idea	Reported result / lesson	Hackathon use
U-Net / Trujillo et al. 2024	Sentinel-1	Encoder-decoder segmentation; Focal/loU losses explored	Up to 96% IoU in their optimised experiment	Baseline
CBD-Net / Zhu et al.	SOS	Contextual + boundary supervision	Designed to improve oil boundaries and context	Stretch if code works
Improved DeepLabV3+	SOS	MobileNetV2 + DeepLabV3+ + scSE	mIoU ~80-81%; F1 ~89% on reported SOS settings	Best stretch
TransUNet implementation	SOS	Transformer encoder + U-Net decoder	Global context; higher complexity	Not for first night
FC-DenseNet + context	Sentinel-1 + contextual data	Segmentation with environmental information	>92% oil-instance detection reported	Research direction
DAB-UNet (2026)	M4D Sentinel-1	Dual attention + boundary-aware multi-class loss	96.19% accuracy, 95.94% F1 reported	Advanced future

Important caution on published scores

Oil-spill papers often use different datasets, splits, preprocessing, class definitions and evaluation protocols. A 96% IoU from one benchmark cannot be directly compared with an 81% mIoU from another without checking the experimental setup. This report therefore uses published results primarily to choose architecture and understand failure modes, not to promise the same score on SIH data.

12. Model Architecture Analysis

12.1 U-Net



```
graph TD; A[Input SAR patch] --> B[Encoder: convolution + downsampling]; B --> C[Bottleneck]; C --> D[Decoder: upsampling + skip connections]; D --> E[1-channel probability map];
```

Input SAR patch
↓
Encoder: convolution + downsampling
↓
Bottleneck
↓
Decoder: upsampling + skip connections
↓
1-channel probability map

The skip connections preserve spatial detail while the encoder captures semantic context. This is exactly the trade-off required for slick segmentation: the model needs to know that a dark region belongs to the ocean context while retaining the boundary of the region.

12.2 DeepLabV3+

DeepLabV3+ combines multi-scale context with a decoder that restores spatial detail. The improved oil-spill study replaced the heavier Xception backbone with MobileNetV2 and used scSE attention. It reported substantially fewer parameters while retaining strong segmentation performance on SOS.

[8]

12.3 Transformer / ViT approaches

TransUNet-style models can capture long-range relationships and global structure. That is attractive for elongated slicks and contextual discrimination, but transformers add implementation and compute overhead. For a one-day build, the main risk is spending time debugging architecture instead of validating the data pipeline.

12.4 Boundary-aware methods

Boundary-aware losses and attention modules are useful when slick edges are weak or fragmented. The 2026 DAB-UNet work combines categorical cross-entropy, Jaccard, Tversky and boundary losses with dual attention, explicitly targeting class imbalance, complex boundaries and look-alike interference.

[15]

13. Recommended Baseline

Baseline: U-Net, one-channel VV input, sigmoid output, BCE + Dice loss.

Component	Choice	Reason
Input	256×256×1 VV	Matches SOS Sentinel-1 subset and minimises complexity
Encoder	Standard U-Net CNN	Fast, proven, easy to debug
Output	256×256×1 sigmoid	Pixel-wise oil probability
Loss	0.5 BCE + 0.5 Dice	Balances pixel accuracy and overlap/class imbalance
Optimizer	Adam	Reliable default
Initial LR	1e-4	Stable starting point
Epochs	20-30	Enough for a first experiment
Metrics	IoU, Dice/F1, precision, recall	Overlap + false-alarm view
Visual QA	Input / GT / prediction triplets	Essential for SAR segmentation

Why BCE + Dice?

BCE penalises pixel-level classification errors, while Dice directly rewards overlap between the predicted and ground-truth regions. Oil can occupy a small fraction of a 256×256 patch, so a loss that only rewards background accuracy can produce deceptively good pixel accuracy while missing the spill. Dice counteracts this tendency.

Stretch model: MobileNetV2-DeepLabV3+. The published oil-spill study used this exact direction and reported roughly 80-81% mIoU and 89% F1 across its SOS experiments, while reducing parameter count substantially compared with Xception-based DeepLabV3+. [8]

Expected difficulty: U-Net 3/10; MobileNetV2-DeepLabV3+ 6/10; TransUNet 8/10; custom physics-informed multi-modal model 9-10/10.

14. Preprocessing and Training Pipeline

14.1 If using SOS (recommended)

```

SOS Sentinel-1 image + ground-truth mask
↓
Read image
↓
Convert / scale to float
↓
Normalise intensity consistently
↓
Ensure 256×256 shape
↓
U-Net
↓
Sigmoid probability
↓
Threshold
↓
Binary oil mask

```

Only include transformations that are consistent between training and inference. Do not add random speckle filters, contrast operations, histogram equalisation or aggressive smoothing simply because they make the image look nicer. Remote-sensing intensity carries physical information; an arbitrary visual enhancement can become a distribution shift.

14.2 If starting from raw Sentinel-1 GRD

```

Sentinel-1 GRD
↓
Radiometric calibration / backscatter product
↓
Land-sea mask
↓
Optional carefully controlled speckle handling
↓
VV / VH selection
↓
Normalisation
↓
Patch extraction
↓
Segmentation model

```

Copernicus provides Sentinel-1 GRD and processing interfaces through the Data Space Ecosystem. The ecosystem also provides Sentinel Hub/openEO-style access for programmatic processing.

[4,21]

14.3 Augmentation

- Horizontal/vertical flips can be reasonable if orientation has no semantic meaning for the model.
- Small rotations can improve robustness, but avoid augmentations that violate acquisition geometry if you later use metadata-aware models.
- Mild intensity noise can simulate variability, but should be tested rather than assumed beneficial.
- Avoid aggressive transformations that turn physical SAR patterns into unrealistic textures.

15. Evaluation Metrics and Validation Design

15.1 Metrics

Metric	Formula / meaning	Why it matters
IoU / Jaccard	$\text{Intersection} / \text{Union}$	Direct overlap measure; strict about false positives
Dice / F1	$2TP / (2TP + FP + FN)$	Useful for imbalanced segmentation
Precision	$TP / (TP + FP)$	How many predicted spill pixels are correct
Recall	$TP / (TP + FN)$	How much actual spill is recovered
Pixel accuracy	$(TP + TN) / \text{all}$	Can be misleading when ocean dominates
Boundary quality	Contour/edge metrics	Important for spill geometry and area estimates

15.2 Why pixel accuracy is dangerous

If 95% of an image is ocean and the model predicts ocean almost everywhere, it can obtain high pixel accuracy while failing the actual task. Therefore, show IoU/Dice plus precision and recall, and always show visual masks.

15.3 Validation leakage

For the SOS benchmark, patch-level random splits can share the same original scene between train and test. A stronger evaluation would split by original scene, event or geographic region. This is especially important if the system is intended for new spills rather than reproducing known benchmark scenes.

15.4 Recommended hackathon reporting

- Report training/validation Dice and IoU.
- Show at least 6-10 qualitative examples: easy, medium, hard and false-positive cases.
- Show at least one intentionally difficult dark look-alike if available.
- State clearly that benchmark performance is not operational accuracy.
- If you add AIS correlation, report it as a downstream attribution layer rather than pretending the segmentation model identifies the polluter itself.

16. From Mask to Spill Characterisation

Once the model returns a binary mask, the next step is not another neural network. Basic geometry is enough to create useful SIH outputs.

Binary mask
↓
Connected-component analysis
↓
Remove tiny isolated components
↓
Largest / relevant slick candidates
↓
Polygonisation
↓
Area, perimeter, centroid, bounding box, elongation

Property	Calculation / interpretation	SIH use
Area	Pixel count × ground area	Estimate affected surface area
Centroid	Mean / weighted centre	AIS search anchor
Bounding box	Min/max coordinates	Quick visualisation
Perimeter	Boundary length	Shape complexity
Aspect ratio	Long axis / short axis	Elongated slick indicator
Orientation	Principal axis angle	Compare with wind/current
Compactness	$4\pi A/P^2$	Distinguish blobs vs filaments

Caution: pixel area is not automatically real-world area unless the image is correctly georeferenced and the pixel ground dimensions are known. For the benchmark, geometry can be demonstrated in pixel coordinates; for real Sentinel-1 GRD, use geospatial metadata and projection-aware area calculations.

A particularly useful output is uncertainty: instead of only a polygon, preserve the mean/maximum model probability and a confidence interval or sensitivity to threshold. This becomes important when the environmental conditions are unfavourable.

17. Environmental Backtracking and Slick Drift

The SIH problem statement goes beyond detection: it envisages using oceanographic and meteorological data to trace a slick toward its origin and predict future movement. ^[1]

17.1 First-order drift model

A simplified conceptual model is:

slick velocity = surface-current component + wind-driven component + wave/Stokes component + unresolved effects

A basic prototype can use the detected centroid at acquisition time and propagate it backward using estimated wind/current vectors. This should be presented as an approximate source-region inference, not a physically complete oil-spill forecast model.

17.2 Why backtracking is uncertain

- Oil does not move exactly with the wind; transport includes current and wave effects.
- The slick can spread and fragment during transport.
- Wind estimates are spatially and temporally averaged.
- Satellite acquisition gives a snapshot, not a continuous movie.
- The observed slick may be older than the most recent vessel activity.
- Different oil types/weathering states can spread differently.

17.3 Hackathon-level implementation

```
Detected centroid + acquisition timestamp
↓
Fetch wind/current vector near the scene
↓
Propagate centroid backward for a few candidate ages
↓
Generate an uncertainty corridor
↓
Use corridor × time window as AIS search region
```

This architecture is attractive because the uncertainty can be passed directly into the AIS module rather than forcing a false point estimate.

18. AIS Correlation and Vessel Attribution

18.1 Concept

AIS provides time-stamped vessel positions, typically including vessel identity and navigational information depending on the source. The attribution problem is therefore a spatiotemporal matching problem: which vessels were close enough to the inferred source region during a plausible time window?

Satellite detection
→ spill polygon + centroid + timestamp
→ environmental backtracking
→ source region + uncertainty + origin window
→ query AIS tracks
→ candidate vessels
→ rank candidates

EMSA explicitly notes that Earth-observation imagery can be combined with AIS and vessel-identification information to support maritime surveillance and identify possible polluters. [5]

18.2 Candidate scoring

Feature	Example score contribution
Distance to inferred origin	Closer = higher
Time consistency	Track intersects plausible origin time = higher
Heading compatibility	Movement direction consistent with slick/source geometry = higher
Speed consistency	Plausible vessel movement during window
Vessel type	Oil tanker / relevant vessel class may receive contextual weight
Track continuity	Continuous track through the region = higher confidence
Environmental consistency	Candidate lies within uncertainty corridor

18.3 Critical data-access caveat

Do not assume the Sentinel-1 mission's onboard AIS data are freely accessible. A current Copernicus Data Space community response states that Sentinel-1C/1D AIS data are restricted to limited authorised users. For SIH, use a separately accessible historical AIS source or a permitted provider; do not build the demo around restricted mission-internal AIS. [23]

19. End-to-End SIH Architecture

```

SATELLITE LAYER
Sentinel-1 SAR image
↓
Quality / land-sea / normalisation
↓
U-Net segmentation
↓
Oil probability mask
↓
Geometry extraction
↓
Spill polygon + centroid + area + confidence

CONTEXT LAYER
Wind / current / acquisition metadata
↓
Environmental quality flag + drift/backtracking
↓
Origin corridor + time window

MARITIME LAYER
AIS trajectories
↓
Spatial-temporal matching
↓
Ranked candidate vessels

```

19.1 Suggested software stack

Layer	Recommended technology
Model	PyTorch
Segmentation	Custom U-Net / torchvision or segmentation_models_pytorch
Raster handling	rasterio, GDAL if needed
Geospatial	GeoPandas, Shapely, pyproj
Visualisation	Matplotlib / Plotly
API/demo	FastAPI + simple web UI, or Streamlit for speed
Satellite access	Copernicus Data Space / Sentinel Hub
Environment	ERA5 or another permitted wind/current source
AIS	Accessible historical AIS provider / dataset

19.2 Minimal system contract

```

Input: Sentinel-1 image + metadata
Output JSON:
{
  spill_detected: true/false,
  confidence: 0.00-1.00,
  centroid: [lat, lon],
  area_km2: ...,
  polygon: ...,
  wind_speed: ...,
  quality_flag: ...,
  candidate_vessels: [...]
}

```

20. Prototype Implementation Plan - Tonight + Tomorrow

The most important rule is do not start with the hardest component. Get a visible mask before adding context or AIS.

Stage	Task	Definition of done
1	Download SOS / refined SOS	Images and masks load in Python
2	Dataset loader	Batch tensor visualises correctly
3	U-Net	Forward pass works
4	Training	Loss decreases and predictions visibly form slick regions
5	Evaluation	IoU/Dice/precision/recall computed
6	Demo visual	Input / GT / prediction panels
7	Geometry	Mask → polygon/centroid/area
8	Wind flag	Scene marked favourable/unfavourable
9	AIS	At least one historical track can be queried and overlaid
10	Stretch	DeepLabV3+ comparison if baseline is stable

Suggested time allocation

- 25%: data + loader + visual inspection
- 35%: U-Net training and debugging
- 15%: evaluation + visual examples
- 10%: geometry extraction
- 10%: wind/context
- 5%: AIS demo integration

If time runs out, stop after Stage 7. A good segmentation demo with honest limitations is better than a broken end-to-end system.

21. Technical Risks and Mitigation

Risk	Severity	Probability	Mitigation
Look-alikes cause false positives	Very high	High	Wind flag + look-alike data + contextual ranking
Benchmark leakage	High	Medium	Document split limitation; scene-level evaluation later
Label noise in SOS	High	Medium	Use refined SOS if feasible
Domain shift to new geography	Very high	High	Treat benchmark as prototype; add diverse data later
Raw Sentinel preprocessing consumes time	High	High	Use preprocessed benchmark first
AIS data unavailable	Medium	Medium	Use permitted historical AIS source; keep module independent
Poor GPU availability	Medium	Medium	Use small U-Net, 256 ² inputs, batch size 4-16
High IoU but poor real-world performance	Very high	High	Qualitative tests + look-alikes + honest reporting
Wind/current model overpromised	High	Medium	Call it approximate source inference, not forensic proof

21.1 Biggest risk: domain shift

Recent research explicitly shows why this matters: datasets with multiple years, regions, morphologies and polarimetric modes can improve real-world generalisation compared with the older SOS benchmark. A recent CDANet study reports that models trained on its broader dataset improved average accuracy by up to 8.3% over models trained on SOS in its real-world evaluation. ^[19]

This is the strongest argument for presenting your system as a candidate detection and attribution assistant, not as a fully autonomous legal/forensic pollution attribution system.

22. Research-Grade Extensions

Once the baseline works, the natural progression is not simply 'bigger model'. It is to make the model more physically aware and less benchmark-dependent.

22.1 Multi-channel SAR

Move from VV-only to VV+VH, potentially adding derived ratios or texture features. The Trujillo dataset demonstrates a practical VV/VH Sentinel-1 segmentation setup. [10]

22.2 Multi-class segmentation

Instead of binary oil/background, predict oil, look-alike, ship, land and sea. The M4D direction is especially attractive because it makes the model explicitly answer the question 'what kind of dark/bright object is this?' rather than only 'is this oil?'. [15]

22.3 Context-aware fusion

Concatenate SAR features with wind speed, wind direction, incidence angle, acquisition time and possibly wave/current variables. This lets the network condition its interpretation on environmental state.

22.4 Temporal modelling

Use multiple Sentinel-1 acquisitions or a temporal encoder. A persistent or evolving dark region can provide stronger evidence than a single snapshot, although changing wind/current conditions complicate direct image differencing.

22.5 Domain adaptation

Recent work on noisy SAR oil-spill segmentation explicitly targets domain shift. This is relevant because a model trained on one geographic distribution can fail on another. Domain-adversarial or feature-alignment approaches can be considered after a strong baseline exists. [20]

22.6 Physics-informed uncertainty

A mature system should lower confidence when wind is outside the useful detection regime, when the predicted object is extremely small, when geometry is implausible, or when the model's probability distribution is uncertain. This is more valuable operationally than squeezing another percentage point out of benchmark IoU.

23. Final Decisions for SIH26143

Question	Final decision
What should we detect?	Candidate oil slicks in Sentinel-1 SAR
Why SAR?	Day/night + cloud-robust + surface-roughness sensitivity
Why Sentinel-1?	C-band, wide-area, open data, mature archive
Primary dataset?	SOS Sentinel-1 subset; consider refined 2025 version
Fallback dataset?	CSIRO if only image-level classification is needed
Robustness dataset?	Trujillo look-alike/no-oil parts, DARTIS, M4D
Baseline?	U-Net + BCE/Dice
Stretch?	MobileNetV2-DeepLabV3+
Preprocessing?	Normalise + shape consistency; avoid unnecessary transforms
Most important context?	Wind speed/direction
Biggest false positive?	Low-wind / natural surface films and other dark look-alikes
Post-processing?	Connected components + polygon + geometry
AIS interface?	Origin corridor + time window → candidate track ranking
Main claim?	Candidate detection + evidence-based attribution, not chemical proof

23.1 The core story for judges

Detection: Sentinel-1 SAR finds dark surface anomalies caused by reduced small-wave backscatter. Discrimination: segmentation plus environmental context reduces obvious false alarms. Characterisation: the mask becomes a geospatial spill object with area, centroid and shape. Attribution: drift/backtracking creates a source corridor, and AIS reconstructs vessels that were plausibly present in the origin window.

23.2 One-line architecture

Sentinel-1 SAR → U-Net → spill mask → geometry → wind/current backtracking → origin corridor → AIS trajectory ranking

24. References and Verified Links

The following sources were used to verify the major claims in this report. Links are included so the team can inspect the original material.

- [1] SIH26143 problem description / research explorer
<https://sih26ps.vercel.app/>
- [2] ESA Sentinel-1 facts and figures - current constellation, C-band, modes, resolution
https://www.esa.int/Applications/Observing_the_Earth/Copernicus/Sentinel-1/Facts_and_figures
- [3] Copernicus Data Space - Sentinel-1 collection and free access
<https://dataspace.copernicus.eu/data-collections/copernicus-sentinel-missions/sentinel-1>
- [4] Copernicus Sentinel-1 documentation
<https://documentation.dataspace.copernicus.eu/Data/Sentinel1.html>
- [5] EMSA - How does SAR detection work?
<https://www.emsa.europa.eu/earth-observation-products/how-does-sar-detection-work.html>
- [6] ESA EO4Society - oil spill SAR training material
https://eo4society.esa.int/wp-content/uploads/2022/01/OCEA03_OilSpill_Kuwait.pdf
- [7] Zhu et al. / SOS dataset description via PMC
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9601461/>
- [8] Improved DeepLabV3+ oil-spill detection
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11397947/>
- [9] Trujillo-Acatitla et al. 2024 - Marine oil spill detection and segmentation
<https://www.sciencedirect.com/science/article/pii/S0025326X24005265>
- [10] Zenodo - Sentinel-1 oil spill dataset Part I
<https://zenodo.org/records/8346860>
- [11] Zenodo - refined SOS dataset
<https://zenodo.org/records/15298010>
- [12] DARTIS dataset paper/preprint
<https://essd.copernicus.org/preprints/essd-2025-208/essd-2025-208-manuscript-version4.pdf>
- [13] PANGAEA - DARTIS dataset
<https://doi.pangaea.de/10.1594/PANGAEA.980773>
- [14] M4D deep neural network oil-spill segmentation paper
https://m4d.iti.gr/wp-content/uploads/2022/12/CNN_Oil_detection_v1.1.pdf
- [15] Boundary-aware dual-attention U-Net / M4D dataset study
<https://pubmed.ncbi.nlm.nih.gov/42097091/>
- [16] TransUNet oil-spill implementation
<https://github.com/Mahesh-Kiran/TransUNET>
- [17] Deep-SAR / CBD-Net project repository
<https://github.com/CUG-URS/Marine-Oil-Spill-Detection-Network-CBDNet>
- [18] FC-DenseNet + contextual information study
<https://arxiv.org/abs/2204.06371>
- [19] Recent CDANet multi-region dataset/work
<https://www.sciencedirect.com/science/article/pii/S0304389425036039>
- [20] Domain-adaptation U-Net oil-spill segmentation
<https://doi.org/10.1038/s41598-026-49934-w>
- [21] Copernicus Data Space - Sentinel Hub service
<https://dataspace.copernicus.eu/ecosystem/services/sentinel-hub>
- [22] ERA5 daily-statistics application documentation
https://dast.data.compute.cci2.ecmwf.int/documents/app-c3s-daily-era5-statistics/C3S_Application-Documentation-ERA5-daily-statistics-v2.pdf
- [23] Copernicus AIS access discussion - restricted Sentinel-1C/D AIS
<https://forum.dataspace.copernicus.eu/t/availability-of-sentinel-1-ais-data/4798>

Appendix A - Quick Dataset Acquisition Checklist

- Download SOS or refined SOS.
- Confirm image and mask filenames align.

- Open 10 random image/mask pairs before training.
- Check image dtype, range and orientation.
- Visualise mask overlay.
- Confirm Sentinel-1 subset is being used for the SIH baseline.
- Keep the original train/test split for reproducibility.
- Record the exact dataset version and download date.

Appendix B - Minimum Demo Screens

- 1. Sentinel-1 SAR input.
- 2. Model probability heatmap.
- 3. Thresholded oil-spill mask.
- 4. Spill polygon with centroid and area.
- 5. Wind-speed / detection-quality indicator.
- 6. AIS vessel trajectories overlaid around the inferred origin corridor.
- 7. Ranked candidate vessel list with distance/time consistency.

Appendix C - What not to claim

- Do not claim SAR proves the material is petroleum.
- Do not claim a benchmark IoU is real-world accuracy.
- Do not claim AIS proximity proves legal responsibility.
- Do not claim a point source location when the drift model only supports an uncertainty corridor.
- Do not hide low/high-wind limitations.

Final technical position

The scientifically defensible prototype is a candidate oil-spill detection and attribution-assistance system. Sentinel-1 provides the all-weather surface-roughness observation; U-Net provides a fast segmentation baseline; wind and ocean context qualify the detection and support approximate backtracking; AIS provides independent behavioural evidence for vessel attribution. The biggest research opportunity after the hackathon is reducing look-alike false alarms and geographic domain shift through multi-class datasets, multi-polarisation inputs, environmental fusion and uncertainty estimation.